INCH-POUND
MIL-DTL-0048684B (AR)
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MIL-C-48684A (AR)
05 December 2001

DETAIL SPECIFICATION

CARTRIDGE, 40MM: M918TP
LOADING, ASSEMBLING, AND PACKAGING

This specification is approved for interim use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC). Other activities in the Department of Defense may use this interim revision or may continue using MIL-C-48684A (AR).

1. SCOPE

1.1 Scope. This specification covers the metal parts and loading, assembling and packaging for one type of cartridge designated as Cartridge, 40MM: M918 TP.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless specified, the issues of these documents are those cited in the solicitation.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000, or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-60942	Primer, Percussion for 40MM, Loading, Assembling and Packing
MIL-A-70625	Automated Acceptance Inspection Equipment Design, Testing and Approval of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	DOD Preferred Method for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

US ARMY (ARDEC) DRAWINGS (see 6.6)

PRODUCT AND PACKAGING DRAWINGS

8886327	Cartridge Case Assembly, 40MM: M169
8886405	Linking of 40MM High Velocity Ammunition
9399372	Cartridge, 40MM: M918TP
9399374	Projectile Assembly
12928042	Packing and Marking of Shipping and Storage Container PA120 with Linked 40MM Cartridges

INSPECTION EQUIPMENT DRAWINGS

9202253	Gage, Chamber, Profile and Alignment 40MM Cartridge
9202254	Gage, Plug, Wear Limit Check
9202255	Gage, Plug, Acceptance Check
9202929	Cartridge Action Time, M129 Grenade Launcher
9208528	Flush Pin Gage
9208529	Flush Pin Gage

(Copies of these drawings may be requested online at drawing-request@pica.army.mil, or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM E1255 - Standard Practice for Radioscopy

(Copies of this document are available online at www.astm.org or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials, parts, and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 X-ray examination of projectile assembly prior to assembly to cartridge case assembly. The projectile assembly shall be X-rayed for improper assembly, missing or damaged parts, foreign material and an armed or partially armed fuze.

3.4 X-ray examination of ballistic sample. Prior to ballistic testing, the cartridge shall be subjected to x-ray examination.

3.5 Functioning. The cartridge assembly shall function, and not cause any weapon stoppages due to the operation of loading, uncoupling, firing and ejecting, and comply with the following:

3.5.1 Velocity. When fired single shot from the MK19, MOD III, Grenade Machine Gun, the mean velocity of the sample cartridge shall be 790 ± 10 feet per second (fps) and a standard deviation not exceeding 12.0 fps.

3.5.2 Security of crimp of base plug and action time. When fired from the M129 Grenade Launcher, the action time of the cartridge shall not exceed 4 milliseconds and there shall be no base plug movement more than .005 above flush with the rear of cartridge case after firing.

3.5.3 Impact functioning. When fired from a MK19, MOD III, Grenade Machine Gun the projectile assembly shall function satisfactorily upon impact as evidenced by smoke, flash and sound. There shall be no evidence of an early burst (see 6.8.1).

3.5.4 Safety. When fired from the MK19, MOD III, Grenade Machine Gun or M129 Grenade Launcher the projectile shall not lodge in the bore of the weapon and there shall be no evidence of a premature burst (see 6.8.2).

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3.6 Ammunition lot numbering. Ammunition Lot Numbering shall be assigned in accordance with MIL-STD-1168.

3.7 Primer functioning.

3.7.1 Primer functioning. The primer shall function without evidence of misfires (see 6.8.3) and in accordance with MIL-P-60942. There shall be no evidence of primer leak as evidenced by perforation of primer cup or smoke deposit on base plug or cartridge case.

3.7.2 Primer deterioration. For deterioration function see paragraph 4.5.3 of MIL-P-60942; furthermore there shall be no evidence of primer leak as evidenced by perforation of primer cup or smoke deposit on base plug or cartridge case.

3.8 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough and workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All marking and stamping shall be neat and sharply defined.

4. VERIFICATION

Table I. Requirements/Inspection Cross Reference Matrix

METHOD OF INSPECTION
N/A - Not applicable
1 - Analysis
2 - Demonstration
3 - Examination
4 - Test

CLASSES OF INSPECTION
A - First article
B - Conformance

Section 3 Requirement	Inspection Methods					Inspection Class		Section 4 Inspection Method
	N/A	1	2	3	4	A	B	
3.1			X	X	X	X		4.3
3.2			X	X	X	X	X	4.3, 4.4, 4.5.1, 4.5.2, 4.5.3, 4.5.6, 4.5.7
3.3				X		X	X	4.5.4
3.4				X		X	X	4.5.5
3.5					X	X	X	4.5.8
3.5.1					X	X	X	4.5.8.1/Table III
3.5.2					X	X	X	4.5.3/Table III
3.5.3					X	X	X	4.5.8/TableIV
3.5.4				X		X	X	4.5.8/4.5.8.2/Table III
3.6				X		X	X	4.5.9
3.7.1					X		X	Table III
3.7.2					X		X	4.5.1.1/Table III
3.8				X		X	X	4.5.11

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4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First article inspection. When specified in the solicitation or contract, a sample shall be subjected to first article inspection in accordance with Table II and 4.3.1.

4.3.1 First article quantity. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provision of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Cartridge Case Assy. M169	8886327	352
Cartridge, 40MM, M918TP	9399372	352
Linked Ammunition, Prior to Packing	12928042	256
Shipping and Storage Container	9362543	11
Projectile Assembly	9399374	352

4.3.2 Inspections to be performed. The first article inspection shall be 100% examination for defects of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed in 4.3.1 and the tests specified in Table II herein.

4.3.3 First article rejection. If any item of the sample fails to comply with the first article requirements, the sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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Table II. First Article Inspection

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
	Projectile and Components			See Below
				NEXT HIGHER ASSEMBLY
EXAMINATION OR TEST		CONFORMANCE CRITERIA/QUANTITY	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Cartridge, 40MM, M918 TP (Dwg. 9399372)				
Functioning Tests		240		
MK19 MOD III – Single Shot		144	3.5	4.4.2.4, 4.5.8.1 TABLE III
MK19 MOD III – Rapid Fire		96 1/	3.5	4.4.2.4, 4.5.8.2
Air Pressure		56	3.2	4.5.8.1
Pull Test		56	3.2	4.5.8.2
Notes: 1/ informational only				

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. For the components identified below, each inspection lot cartridges shall be produced with components complying with the following constraints:

a. M169 cartridge case loading assemblies metal parts from one interfix number, from one manufacturer.
b. Primers from one lot number from one manufacturer.
c. Projectile assemblies from no more than two (2) consecutive lots from one lot interfix number and from one manufacturer.
d. Propellant from one lot number from one manufacturer.
e. Only cartridges from one lot per ammunition belt shall be used. No short belts permitted.
f. Links from one interfix lot number from one manufacturer.
g. Metal parts from one lot interfix number, from one manufacturer.

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Verification Level IV shall be used for all characteristics defined as Majors and Verification Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 through 4.4.2.8. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 8886327
4.4.2.1	Cartridge Case Assembly Prior to Loading			NEXT HIGHER ASSEMBLY 9399372
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Closing cup missing	100%	3.2	AAIE/4.5.10
2.	Three (3) or more vent holes blocked or missing 2/	100%	3.2	AAIE/4.5.10
MAJOR				
101.	One (1) or two (2) vent holes blocked or missing	100%	3.2	AAIE/4.5.10
102.	Two or more closing cups 1/	100%	3.2	AAIE/4.5.10/ Radiography
MINOR				
201.	Evidence of poor workmanship	Level II	3.11	Visual/4.5.11

Notes:
1/ May be inspected at higher-level assembly.
2/ Acceptance Inspection of the cartridge case is performed with AIE by the case manufacturer. Verification at the LAP facility is required.

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Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 8886327
4.4.2.2	Cartridge Case Assembly Prior to Assembling Base Plug and Primer			NEXT HIGHER ASSEMBLY 9399372
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Propellant charge weight less than 75% or more than 125% of assessed weight	100%	3.2	AAIE/4.5.1/ 4.5.2
MAJOR				
101.	Propellant charge weight out of drawing tolerance but within +/- 25% of assessed weight	100%	3.2	AAIE/4.5.1/ 4.5.2
MINOR				
201.	Evidence of poor workmanship	Level II	3.11	Visual/4.5.11
Notes:				

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Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 8886327
4.4.2.3	Cartridge Case Assembly			NEXT HIGHER ASSEMBLY 9399372
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Primer above flush with base plug	100%	3.2	AAIE/4.5.10
MAJOR				
101.	Security of crimp of base plug and action time	80 2/	3.5.2	4.5.3
102.	Base plug above flush	Level IV	3.2	9202528
103.	Depth of base plug, max	Level IV	3.2	9202528
104.	Depth of primer from base plug, max	Level IV	3.2	9202529
105.	Primer damaged	Level IV	3.2	Visual
106.	Excessive sealing compound on exterior surface of primer	Level IV	3.2	Visual
107.	Deterioration of primers	1/	3.7.2	4.5.1.1
MINOR				
201.	Evidence of poor workmanship	Level II	3.8	Visual/4.5.11

Notes:
1/ Sampling and rejection shall be in accordance with MIL-P-60942.
2/ Accept on zero (0) and reject on one (1).

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Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399374
4.4.2.4	Projectile Assembly			NEXT HIGHER ASSEMBLY 9399372
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	X-Ray examination 1/	100%	3.4	4.5.5
MAJOR				
101.	X-Ray examination 2/	100%	3.4	4.5.5
102.	O-ring missing from groove	Level IV	3.2	Visual/9399372
MINOR				
201.	Evidence of poor workmanship	Level II	3.8	Visual/4.5.11

Notes:

1/ Critical Defect is defined as (for definitions see MIL-DTL-48669):

a. Rotor in armed or partially armed position
b. Detent missing
c. Setback spring missing
d. Setback pin missing or retracted
e. Pinion or verge assembly missing
f. Detent not engaged in teeth of sector gear
g. Setback pin not blocking rotor
h. Detent spring missing

2/ Major Defect is defined as (for definitions see MIL-DTL-48669):

a. Foreign material present.

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Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399372
4.4.2.5	Cartridge, 40MM. M918TP			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	X-ray examination of ballistic sample 1/	240	3.5	4.5.5
MAJOR				
101.	Chamber gage failure	100%	3.2	Gage, 9202253/ 9202254/ 9202255
102.	Total length, max	100%	3.2	Gage
103.	X-ray examination of ballistic sample 2/	240 3/	3.5	4.5.5
104.	Air pressure test	Level III	3.2	4.5.6
105.	Pull test of projectile	Level III	3.2	4.5.7
106.	Gap, max, at rear of rotating band	Level IV	3.2	Gage
107.	Packing exposed	Level IV	3.2	Visual
	Functioning Tests	240	3.5	
108.	MK19 MOD III Machine gun – Single shot	144 4/	3.5/3.5.1 3.5.3/3.5.4	4.5.8.1
109.	MK19 MOD III Machine gun – Rapid fire	96 4/	3.5/3.5.3 3.5.4	4.5.8.2
MINOR				
201.	Marking missing, improper or illegible	Level II	3.2	Visual
202.	Evidence of poor workmanship	Level II	3.8	Visual/4.5.11

Notes:
1/ A critical defect is defined as any defect listed in Note 1 of 4.4.2.4 plus a low or high propellant charge or a missing closing cup. Accept on 0 reject on 1.
2/ A major non-conformance is defined as any non-conformance specified on note 2 of 4.4.2.4 or a minor non-conformance.
3/ Any major non-conformance found shall be reported with the exception of two or more closing cups which shall result in the withholding of acceptance of the lot and ballistic sample.
4/ See Table III for defect classification and acceptance/rejection criteria.

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Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 8886405
4.4.2.6	Linked Ammunition, Prior to Packing			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Linked improperly assembled (anyone of the four (4) tabs of link not in groove, link inverted, link damaged, link distorted)	100%	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.8	Visual/4.5.11
Notes:				

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Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 12928042
4.4.2.7	Shipping and Storage Container (unsealed)			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Cartridge inverted	Level IV	3.2	Visual
102.	Incorrect number of cartridges	Level IV	3.2	Visual
103.	Fillers missing	Level IV	3.2	Visual
104.	Support, top and bottom missing (when applicable)	Level IV	3.2	Visual
105.	Spacer missing	Level IV	3.2	Visual
106.	Top pad missing (when applicable)	Level IV	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.8	Visual/4.5.11

Notes:

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Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 12928042
4.4.2.8	Shipping and Storage Container (sealed)			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Box or container damaged	Level IV	3.2	Visual
MINOR				
201.	Contents loose	Level II	3.2	Manual
202.	Can seal missing or improperly positioned	Level II	3.2	Visual
203.	Marking incorrect or illegible	Level II	3.2	Visual
204.	Evidence of poor workmanship	Level II	3.8	Visual/4.5.11

Notes:

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TABLE III. Conformance Inspection Functioning Defect Classification and Acceptance / Rejection Criteria

	Acceptance number		Rejection Number		Classification Defect
	MK19 Single Shot 4/	MK19 Rapid Fire	MK19 Single Shot 4/	MK19 Rapid Fire	
1. Failure to meet mean velocity (see 3.5.1)	1/	--	2/	--	Major
2. Standard deviation of velocity over max (see 3.5.1)	1/	--	2/	--	Major
3. Impact functioning failure (see 3.5.3) 5/	8/(5)	3/	9/(6)	3/	Major
4. Primer leak as evidenced by perforation of primer cup or smoke deposit on base plug or cartridge case (see 3.7.2)	0	0	1	1	Major
5. Projectile lodges in weapon bore (see 3.5.4)	0	0	1	1	Critical
6. Premature Burst (see 3.5.4)	0	0	1	1	Critical
7. Early Burst (see 3.5.3)	0	0	1	1	Major
8. Primer Misfire (see 3.7.1)	0	0	0	1	Major
9. Weapon stoppage excluding primer misfire (see 3.5)	0	0	1	1	Major

Notes:
1/ Lot acceptable if requirement is met.
2/ Lot withheld from acceptance if requirement not met.
3/ For informational purposes only.
4/ M129 Grenade Launcher is used to measure Action Time & Security of crimp of base plug.
5/ First Article Accept/Reject number are indicated in the parentheses

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4.5 Methods of inspections.

4.5.1 Primer

4.5.1.1 Check test for deterioration of primers. If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions, however, brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-60942 immediately before the primer lot is assembled into the cartridge. The test shall be performed at the facility assembling the primers into the cartridge.

4.5.2 Propellant weight. The propellant quantity shall be determined and then weighed 100 percent. Any charge that fails to comply with the drawing requirement (taking into account machine accuracy) shall be classified defective and removed from the lot. The accuracy of the machine shall be determined and the prove-out test performed to validate the machine's ability to remove charges outside of the drawing tolerance range. The machine must be capable of detecting and rejecting critical propellant weights, such that the end-product contains less than 1 in a million critical defects (examples machine error rate not to exceed 1/500 and maximum defect rate not to exceed 0.05 percent).

4.5.3 Security of crimp of base plug and action time. The cartridge assemblies shall be assembled to a projectile and fired from a M129 Grenade Launcher. After firing, observation shall be made for base plug movement and action time. If any base plug movement or action time exceeds the applicable requirement, or any projectile lodges in the bore the quantity represented shall be rejected and set aside for disposition in accordance with MIL-STD-1916. The movement of the base plug shall be measured from the highest point on the base plug to a point on the cartridge case directly beside it. The cartridge case assemblies produced by each primer assembly machine shall be kept segregated and identified. They shall not be used in production until successful completion of this test.

4.5.4 X-ray or fluoroscopic examination of projectile assemblies. Prior to assembly to the cartridge case assembly, the projectile assembly shall be inspected for critical and major defects using approved x-ray or fluoroscopic equipment. The test shall be conducted in two (2) planes. One shall be from the top of the projectile down and the other from the side. The test shall be conducted in accordance with ASTM E1255.

4.5.5 X-ray examination of ballistic sample. Prior to ballistic testing, the cartridges shall be inspected for defects using approved x-ray equipment. This test shall be conducted in two planes: from the top of the projectile down and from the side. If any critical defect is found, the lot shall be rejected including the ballistic sample. The test shall be conducted in accordance with ASTM E1255.

4.5.6 Air pressure. The cartridge assembly shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Observations shall be made for a drop in pressure indicating a leak. This is a non-destructive test. Parts successfully tested may be returned to the lot.

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4.5.7 Pull test of projectile. The cartridge shall be placed in an approved fixture and the axial tension specified on the applicable drawing shall be applied. If the sample fails the drawing requirement, the lot represented by the sample shall be rejected and set aside for disposition in accordance with MIL-STD-1916, root cause determined and corrective action taken. Cartridge case assemblies may be reused after undergoing a Government approved rework provided the same crimping method is used. Projectile assemblies if undamaged may be reused. The cartridge shall be pulled until separation occurs and the data analyzed.

WARNING: This test is conducted on assemblies containing an explosive element. Test barricades (where used), procedures and equipment shall have prior Government approval.

4.5.8 Functioning tests. The tests shall be conducted at a Government approved test facility. If new, the MK19, MOD III, Grenade Machine Gun and barrel shall have at least forty (40) rounds fired on it prior to being used for the test. The distance to the impact medium of coarse sand shall be 200 meters plus or minus 5 meters. The depth of the sand shall be a minimum of 10 inches.

4.5.8.1 Single shot. The barrel selected for the test must be used throughout the entire test. The rounds shall be conditioned at 70 ± 5º F for at least 8 hours and then fired within 15 minutes of being removed from conditioning temperature. The cartridges shall be fired single shot from a MK19, MOD III, Grenade Machine Gun at a rate not to exceed ten (10) per minute. At least six (6) gun warmers shall be fired at the beginning of the test. All six (6) warmers shall be fired within one (1) minute. Rake sand prior to beginning of the test (each ammunition lot). Rake sand as required such that individual impact craters do not overlap. Sand shall be raked after each function failure. If a round falls outside the specified impact area, it shall be declared a “no test” for impact functioning requirements but velocity measures shall still be used. The velocity shall be measured at 30 feet from the muzzle of the launcher. If an approved radar system is not used, then the monitoring devices and time recorders shall be a dual system setup for recording two (2) separate readings that shall be within 3 feet per second of each other. If the two readings differ by more than 3 feet per second (for a given shot), the results shall be declared a no test and another round fired in its place. The number of “no test” rounds shall be reported. “No test” rounds shall be reported as “no test” rounds outside the impact area or “no test” for velocity reading. Report the approximate range to impact of “no test” rounds. Results shall be calculated and recorded for mean velocity and standard deviation (see 6.7). Observation shall be made for compliance with Table III.

4.5.8.2 Rapid fire. The cartridges shall be rapid fired from a MK19, MOD III, Grenade Machine Gun mounted on the MK64 Mount and M3 Tripod. Firing shall be in bursts of five (5) to ten (10) rounds. Rate of fire shall be documented. Observation shall be made for compliance with Table III.

4.5.9 Verification of ammunition lot numbering. Visually verify that an ammunition lot number has been applied to each lot as described in MIL.-STD-1168.

4.5.10 Inspection Equipment. The inspection equipment required to perform the

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examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. Any automated acceptance inspection equipment (AAIE) prove-out shall be performed in accordance with MIL-A-70625 (See 6.10).

4.5.11 Verification of workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.8.

5. PACKAGING

5.1 For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the metal parts and loading, assembling and packing for Cartridge, 40MM: TP, M918. The Target Practice (TP) designation for flash, smoke, and sound, makes the item military unique.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number (w/revision level) and date of all reference documents.
c. Packaging requirements.
d. Requirements for First Article (see 4.3).
e. Government Furnished Material or equipment (if applicable).
f. Requirements for acceptance inspection equipment (AIE) designs (see 6.10).
g. Requirements for Ammunition Lot Numbering (6.11).
h. Applicable national stock number.
i. Serialization requirements, if applicable.
j. Certificate of conformance for each lot or shipment of product.
k. Applicability of Live Fire T&E requirements under Public Law 1033-355, Section 2366 of Title 10, USC.
l. Information needed to satisfy the requirements for qualification of energetic materials by the Army Service Qualification Authority, (see 6.12). This information can be obtained in the desktop guide titled

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"Explosive Hazard Classification Requirements for Performance Specifications", Revision 2, dtd 20 May 1998 that is available from AMSRD-AAR QES-C.

6.3 Submission of results of contractor conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor should forward the requested records of examination or tests to the PCO at JMC. Copies may be provided to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000.

6.4 Subject term (keyword) listing.

M550 Escapement
MK19 MOD III Grenade Machine Gun
PA120 Shipping and Storage Container

6.5 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the US Army Armament Command, one (1) copy of all ballistic data and ammunition data cards should be forwarded to the PCO at JMC. Copies may be provided to: Commander, US Army RDECOM-ARDEC ATTN: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM),.Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of RDECOM-ARDEC.

6.7 Standard deviation should be calculated from the following formula:

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Where: x = each individual value

$\bar{x}$ = sample arithmetic mean $\quad \frac{\sum x}{n}$

n = sample size

$\sum (x - \bar{x})^2$ = The sum of the squares of the differences between the sample mean and each individual value.

6.7.1 Alternate method. For a faster and easier method of arriving at the standard deviation formula, the following may be used:

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$$S = \sqrt{\frac{n\sum x^2 - (\sum x)^2}{n(n-1)}}$$

6.8 Definitions.

6.8.1 Early burst. A projectile that functions in flight further than the arming distance from the weapon muzzle.

6.8.2 Premature burst. A premature burst is considered to have occurred if the projectile functions in the weapon bore or in flight up to the arming distance from the weapon muzzle.

6.8.3 Misfire. Failure of a round of ammunition to fire after the initiating impulse has been applied to the primer or the failure of the initiated primer to ignite the propellant.

6.9 Submission of alternative conformance provisions. All contractor proposed alternative conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity

6.10 Critical Inspection Equipment.

6.10.1 Contractor acceptance inspection equipment (AIE). Provisions concerning the contractor's AIE used to verify the requirements of this specification should be specified in the contract.

6.10.2 System Reliability. The contractor should provide for each defect identified as Critical, a manufacturing and inspection system that assures no more than one in a million contain the defect. This should be assured by controlling the maximum defect rate produced and the error rate of the inspection equipment such that the product of the two terms is less than one in a million.

6.10.3 Maximum defect rate produced. The maximum defect rate produced is defined as the largest defect rate expected for the characteristics of concern on a monthly or lot basis. The contractor should establish the maximum defect rate, however, it may not exceed one percent without approval of the procuring contracting officer. Once established, the contractor should monitor the defect rate to assure it does not exceed the maximum rate allotted. If the established maximum defect rate is exceeded, the production of that feature should be terminated until the cause is determined and corrected. Additionally, all products for that period of time should be rejected and re-inspected.

6.10.4 Maximum error rate of the inspection system. The error rate of the inspection equipment should be defined as the expected ratio of the number of defective parts accepted to the number of defective parts inspected by the equipment. The

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contractor, to meet the system reliability as defined in contract, defines the maximum error rate allowed. However, it may not exceed 1/500 without approval of the procuring contracting officer. Based on the maximum error rate defined for the equipment, the contractor should develop demonstration test procedure to demonstrate the error rate of the equipment. The test is performed using defective parts or reject standards. No part or standard should be accepted during the test. If a part or standard is accepted, the cause for failure should be isolated and corrected and the test rerun. Unless otherwise specified in the detailed item specification, the minimum number of test samples to be run should be equal to 0.7 divided by the error rate (e.g. If the expected error rate is 1/1000, the sample size would be 0.7 divided by (1/1000) = 700.).

6.10.5 Periodic verification. Periodic verification of the system error rate should be performed.

6.10.6 Redundant inspection equipment. The contractor may elect to use redundant inspection equipment in lieu of using a single piece of equipment with a very low error rate. Where redundant equipment is used, the inspection system error rate is defined as the error rates of the multiple pieces of equipment multiplied together.

6.11 Ammunition lot numbers. Ammunition requires ammunition data cards in accordance with MIL-STD-1168.

6.12 Energetic material qualification information. Qualification of energetic materials by the Army Service Qualification Authority requires the contractor to demonstrate that the energetic material meets the requirements of MIL-STD-1751 or STANAG 4170. This task is addressed in the statement of work in the contract. Point of contact for the Army Service Qualification Authority can be obtained from AMSRD-AAR-QES-C.

6.13 Revisions/Amendments History. The following is a record of changes incorporated into this document

ECP	Description	Date Approved
R04Q2018	Initial release of M918 Cartridge MIL-DTL Specification	Aug 2004

6.14 Interim Revision This interim revision has been prepared at Program Manager request to document in ASSIST Online the version of a draft specification that was placed on contract in 2004. This interim revision has been dated to match the date of the draft specification that was placed on contract. It has been modified slightly to comply with MIL-STD-961 requirements for format, but not content. A fully coordinated revision, compliant with both MIL-STD-961 format and content requirements, will be approved for publication.

6.15 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Army-AR
(Project No. 1310-2006-010)

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